Math. Z. (2007) 255:95–115
DOI 10.1007/s00209-006-0014-2

Mathematische Zeitschrift

The *L*-functions of Witt coverings

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Received: 21 March 2005 / Accepted: 18 August 2005 /
Published online: 11 July 2006
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Abstract Results on L-functions of Artin–Schreier coverings by Dwork, Bombieri and Adolphson–Sperber are generalized to L-functions of Witt coverings.

Keywords L-functions · Exponential sums · Newton polygon

Mathematics Subject Classification (2000) 11L07 · 14F30

1 Introduction

We shall state our main results after recalling the notion of L-functions of Witt coverings. Let $\mathbb{F}_q$ be the finite field of characteristic p with q elements, and W_m the ring scheme of Witt vectors of length m over $\mathbb{F}_q$. Let $f \in W_m(\mathbb{F}_q[x_1^{\pm 1}, \ldots, x_n^{\pm 1}])$ with its first coordinate non-constant. Let T^n be the n-dimensional torus over $\mathbb{F}_q$, and F the Frobenius morphism of W_m. The fibre product over W_m of $W_m \stackrel{F-1}{\to} W_m$ and $T^n \stackrel{f}{\to} W_m$ is a $W_m(\mathbb{F}_p)$-covering of T^n, with group action $g(y, x) = (y + g, x)$. The Frobenius element of the Galois group $W_m(\mathbb{F}_p)$ at a closed x of T^n with degree k is $\mathrm{Tr}_{W_m(\mathbb{F}_{q^k})/W_m(\mathbb{F}_p)}(f(x))$. So the Artin L-function of T^n determined by that $W_m(\mathbb{F}_p)$-covering and a fixed character $\psi : W_m(\mathbb{F}_p) \to \overline{\mathbb{Q}}^{\times}$ of exact order p^m is

$$L_f(t) = \prod_{x \in |T^n|} (1 - \psi(\mathrm{Tr}_{W_m(\mathbb{F}_{q^{\deg(x)}})/W_m(\mathbb{F}_p)}(f(x)))t)^{(-1)^n},$$

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where $|T^n|$ is the set of closed points of T^n. By a well known theorem of Deligne [5],

$$L_f(t) = \frac{\prod_\alpha (1-\alpha t)}{\prod_\beta (1-\beta t)},$$

where α and β are algebraic integers such that $q^n\alpha^{-1}$ and $q^n\beta^{-1}$ are also algebraic integers. It implies, as observed by Bombieri [4], $\mathrm{ord}_q(\alpha), \mathrm{ord}_q(\beta) \le n$, where ord_q is the q-order function of $\overline{\mathbb{Q}}_p$ such that $\mathrm{ord}_q(q) = 1$. ($\overline{\mathbb{Q}}_p$ is the algebraic closure of $\mathbb{Q}_p$, the field of p-adic numbers.) By logarithmic differentiation, we get

$$L_f(t) = \exp\left(\sum_{k=1}^{\infty} S_k(f)\frac{t^k}{k}\right),$$

where

$$S_k(f) = (-1)^{n-1} \sum_{x\in(\mathbb{F}_{q^k}^\times)^n} \psi(\mathrm{Tr}_{W_m(\mathbb{F}_{q^k})/W_m(\mathbb{F}_p)}(f(x)))$$

are exponential sums associated to characters of p-power order. To have a look at these exponential sums, we denote by $\lambda_i : A^1 \to W_m$, $i = 0, \ldots, m-1$, the embedding which maps A^1 onto the i-th axis of W_m, and write

$$f = \sum_{i=0}^{m-1} \sum_{u\in I_i} \lambda_i(a_{iu}x^u),$$

where $I_i \subset \mathbb{Z}^n$ and $a_{iu} \in \mathbb{F}_q^\times$ are uniquely determined. That decomposition can be obtained by solving the congruences

$$\begin{aligned}
f &\equiv \lambda_0\left(\sum_u a_{0u}x^u\right) (\mathrm{mod}\ V)\\
f - \sum_u \lambda_0(a_{0u}x^u) &\equiv \lambda_1\left(\sum_u a_{1u}x^u\right) (\mathrm{mod}\ V^2)\\
&\vdots\\
f - \sum_{i=0}^{m-2}\sum_u \lambda_i(a_{0u}x^u) &\equiv \lambda_{m-1}\left(\sum_u a_{(m-1)u}x^u\right) (\mathrm{mod}\ V^m)
\end{aligned}$$

successively, where V is the shift operator on W_m.

Let $\overline{\mathbb{F}}_q$ be the algebraic closure of $\mathbb{F}_q$, and ω the Teichmüller lifting from $\overline{\mathbb{F}}_q$ to $\overline{\mathbb{Q}}_p$. We define $\omega(f) = \sum_{i=0}^{m-1} p^i \sum_{u\in I_i} \omega(a_{iu})x^u$. Let $\mathbb{Z}_p$ be the ring of p-adic integers, and μ_l ($l \ge 1$) be the set of l-th roots of unity in $\overline{\mathbb{Q}}_p$. Identifying $W_m(\mathbb{F}_{q^k})$ with $\mathbb{Z}_p[\mu_{q^k-1}]/(p^m)$ under the isomorphism

$$(a_0, \ldots, a_{m-1}) \mapsto \sum_{j=0}^{m-1} \omega\left(a_i^{p^{-i}}\right) p^i \ (\mathrm{mod} p^m),$$

one finds, for $x \in (\mathbb{F}_{q^k}^\times)^n$, that

$$\begin{aligned}\psi\left(\mathrm{Tr}_{W_m(\mathbb{F}_{q^k})/W_m(\mathbb{F}_p)}(f(x))\right) &= \psi\left(\mathrm{Tr}_{\mathbb{Q}_p[\mu_{q^k-1}]/\mathbb{Q}_p}\left(\sum_{i=0}^{m-1}\sum_u p^i\omega\left(a_{iu}^{p^{-i}}x^{p^{-i}u}\right)\right)\right)\\ &= \psi\left(\mathrm{Tr}_{\mathbb{Q}_p[\mu_{q^k-1}]/\mathbb{Q}_p}\left(\sum_{i=0}^{m-1}\sum_u p^i\omega\left(a_{iu}x^u\right)\right)\right)\\ &= \psi\left(\mathrm{Tr}_{\mathbb{Q}_p[\mu_{q^k-1}]/\mathbb{Q}_p}(\omega(f)(\omega(x)))\right).\end{aligned}$$

Therefore, we have

Lemma 1.1 *For $k = 1, 2, \ldots$, we have*

$$S_k(f) = (-1)^{n-1}\sum_{x\in\mu_{q^k-1}^n}\psi(Tr_{\mathbb{Q}_p[\mu_{q^k-1}]/\mathbb{Q}_p}(\omega(f)(x))).$$

We define the Newton polyhedron $\Delta_\infty(f)$ of f at infinity to be the convex hull in $\mathbb{Q}^n$ of $\{p^{m-i-1}u : 0 \le i \le m-1, u \in I_i\} \cup \{0\}$. Recall that, for a convex polyhedron Δ of dimension n in $\mathbb{Q}^n$ that contains the origin, there is a $\mathbb{R}_{\ge 0}$-linear degree function $u \mapsto \deg(u)$ on $L(\Delta)$, the set of integral points in the cone $\bigcup_{k=1}^\infty k\Delta$, such that $\deg(u) = 1$ when u lies on a face of Δ that does not contain the origin. That degree function may take on non-integral values. But there is a positive integer D such that $\deg L(\Delta) \subset D^{-1}\mathbb{Z}$. We denote the least positive integer with this property by $D(\Delta)$. For $k = 0, 1, \ldots$, we denote by $W_\Delta(k)$ the number of points of degree $\frac{k}{D(\Delta)}$ in $L(\Delta)$. We define $P_\Delta(t) = (1-t^{D(\Delta)})^n\sum_{k=0}^{+\infty} W_\Delta(k)t^k$ for later use. Our first result is an upper bound for the total degree of $L_f(t)$.

Theorem 1.2 *The total degree of $L_f(t)$ is bounded by $\sum_{i=0}^n \binom{n}{i}\sum_{k=0}^{D(n-i+1)} W_\Delta(k)$ with $D = D(\Delta)$ and $\Delta = \Delta_\infty(f)$.*

For $j = 1, \ldots, n$, we write

$$\overline{{}_jf}^\tau = \sum_{i=0}^{m-1}\sum_{p^{m-i-1}u\in\tau} u_j a_{iu}^{p^{m-i-1}} x^{\,p^{m-i-1}u},$$

where u_j is the j-th coordinate of u. We call f non-degenerate with respect to $\Delta_\infty(f)$ if $\Delta_\infty(f)$ is of dimension n, and for every face τ of $\Delta_\infty(f)$ that does not contain 0, the system $\overline{{}_1f}^\tau = \cdots = \overline{{}_nf}^\tau$ has no common solution in $(\overline{\mathbb{F}}_q^\times)^n$. Our second result is on L-functions from non-degenerate Witt vectors.

Theorem 1.3 *Suppose that f is non-degenerate with respect to $\Delta := \Delta_\infty(f)$. Then the L-function $L_f(t)$ is a polynomial, and its Newton polygon with respect to ord_q lies above the Hodge polygon of $P_\Delta(t)$ of degree $D(\Delta)$ with the same endpoints. In particular, $L_f(t)$ is of degree $n!Vol(\Delta)$.*

Recall that the Newton polygon of $\prod(1-\alpha t) \in \overline{\mathbb{Q}}_p[[t]]$ with respect to ord_q is the polygon with vertices at points

$$\left(\sum_{\mathrm{ord}_q(\alpha)\le y} 1, \sum_{\mathrm{ord}_q(\alpha)\le y}\mathrm{ord}_q(\alpha)\right), \quad y \in \mathbb{Q}.$$

And the Hodge polygon of $\sum_{k=0}^{+\infty} a_k t^k$ of degree D is the polygon with vertices at the points $(0,0)$ and

$$\left(\sum_{i=0}^{k} a_i, \frac{1}{D}\sum_{i=0}^{k} i a_i\right), \quad k = 0, 1, \ldots.$$

When f is non-degenerate, by Theorem 1.3 and a lemma of Kouchnirenko [10, Lemma 2.9], $L_f(t)$ is a polynomial of degree

$$n!\mathrm{Vol}(\Delta) = P_\Delta(1) = \sum_{i=0}^{n}(-1)^i \binom{n}{i} \sum_{k=0}^{D(n-i+1)} W_\Delta(k),$$

which is much sharper than the bound in Theorem 1.2. The bound in Theorem 1.2 governs the worst cases. Fortunately, the first author [13] shows in a paper in preparation that f is "often" non-degenerate. In that paper he also determines the Newton polygon of Gauss–Heilbronn sums.

Theorem 1.3 was proved by Dwork [6] when $m = 1$, and $f(x_1, \ldots, x_n) = x_n h(x_1, \ldots, x_{n-1})$, where h is a polynomial with coefficients in $\mathbb{F}_q$ whose Newton polyhedron at infinity is homothetic to the standart simplex. In that case, the L-function $L_f(t)$, by the orthogonality of characters, is related to the zeta function of the hypersurface defined by $h = 0$ in the $(n-1)$-dimensional affine space defined over $\mathbb{F}_q$. It was completely proved by Adolphson and Sperber [2] in the case $m = 1$. In the case $n = 1$, the degree of $L_f(t)$ was determined by Kumar et al. [11] with applications to coding theory, and by Li [12], who read the $p = 2$ version of [11].

Our proof of the main results is based on the p-adic method set up by Dwork [6,7] and developed by Bombieri [3,4], Monsky [14], Adolphson and Sperber [1,2], Wan [16], and others. The innovation lies in the use of the Artin–Hasse exponential series to produce roots of unity of p-power order. Some different ideas are used in studying the Newton polygons of these L-functions. One can infer the following theorem from Theorem 1.3.

Theorem 1.4 *If f is non-degenerate with respect to $\Delta_\infty(f)$, and the origin lies in the interior of $\Delta_\infty(f)$, then the reciprocal roots of $L_f(t)$ are of absolue value $q^{n/2}$.*

2 The Artin–Hasse exponential series

Let

$$E(t) = \exp\left(\sum_{i=0}^{\infty} \frac{t^{p^i}}{p^i}\right) \in \mathbb{Z}_p[[t]]$$

be the Artin–Hasse exponential series. We shall use it to produce roots of unity of p-power order.

Lemma 2.1 *If l is a positive integer, and π is a root of $\sum_{i=0}^{\infty} t^{p^i}/p^i = 0$ in $\overline{\mathbb{Q}}_p$ with order $1/(p^{l-1}(p-1))$, $E(\pi)$ is a primitive p^l-th root of unity.*

Proof First, $\exp(p^l\pi^{p^i}/p^i)$ exists as $\mathrm{ord}_p(p^l\pi^{p^i}/p^i) \geq p/(p-1)$. So

$$E(\pi)^{p^l} = E(p^l t)|_{t=\pi} = \prod_{i=0}^{\infty} \exp\left(p^l \frac{\pi^{p^i}}{p^i}\right) = \exp\left(\sum_{i=0}^{\infty} p^l \frac{\pi^{p^i}}{p^i}\right) = \exp(0) = 1.$$

Secondly, as $E(t) \in 1 + t + t^2\mathbb{Z}_p[[t]]$,

$$E(\pi)^{p^{l-1}} \equiv (1+\pi)^{p^{l-1}} \equiv 1 + \pi^{p^{l-1}} \left(\bmod \pi^{p^{l-1}+1}\right).$$

The lemma is proved.

Lemma 2.2 *Let l be a positive integer. Then the Artin–Hasse exponential series induces a bijection $\pi \mapsto E(\pi)$ from the set of roots of $\sum_{i=0}^{\infty} t^{p^i}/p^i = 0$ in $\overline{\mathbb{Q}}_p$ with order $1/(p^{l-1}(p-1))$ to the set of all primitive p^l-th roots of unity in $\overline{\mathbb{Q}}_p$.*

Proof The field generated over $\mathbb{Q}_p$ by the set of roots of $\sum_{i=0}^{\infty} t^{p^i}/p^i = 0$ in $\overline{\mathbb{Q}}_p$ with order $1/(p^{l-1}(p-1))$ is precisely $\mathbb{Q}_p(\mu_{p^l})$ since it contains $\mathbb{Q}_p(\mu_{p^l})$ by the preceding lemma, and is of degree no greater than $p^{l-1}(p-1)$ over $\mathbb{Q}_p$ by Weierstrass' Preparation Theorem. One sees that $E(\tau(\pi)) = \tau(E(\pi))$ if τ is an automorphism $\mathbb{Q}_p(\mu_{p^l})$ over $\mathbb{Q}_p$. So $\pi \mapsto E(\pi)$ maps the set of roots of $\sum_{i=0}^{\infty} t^{p^i}/p^i = 0$ in $\overline{\mathbb{Q}}_p$ with order $1/(p^{l-1}(p-1))$ onto the set of all primitive p^l-th roots of unity in $\overline{\mathbb{Q}}_p$. It is a bijection as $\sum_{i=0}^{\infty} t^{p^i}/p^i = 0$ has at most $p^{l-1}(p-1)$ roots in $\overline{\mathbb{Q}}_p$ with order $1/(p^{l-1}(p-1))$ by Weierstrass' Preparation Theorem.

Lemma 2.3 *If k is a positive integer, and $x \in \overline{\mathbb{Q}}_p$ satisfies $x^{p^k} = x$, then*

$$E(t)^{x+x^p+\cdots+x^{p^{k-1}}} = E(tx)E(tx^p)\cdots E\left(tx^{p^{k-1}}\right).$$

Proof As $\sum_{j=0}^{k-1} x^{p^j} = \sum_{j=0}^{k-1} x^{p^{j+i}}$, we have

$$\begin{aligned} E(t)^{x+x^p+\cdots+x^{p^{k-1}}} &= \exp\left(\sum_{i=0}^{\infty} \frac{t^{p^i}}{p^i} \sum_{j=0}^{k-1} x^{p^j}\right) = \exp\left(\sum_{i=0}^{\infty} \frac{t^{p^i}}{p^i} \sum_{j=0}^{k-1} x^{p^{j+i}}\right) \\ &= \exp\left(\sum_{j=0}^{k-1} \sum_{i=0}^{\infty} \frac{(tx^{p^j})^{p^i}}{p^i}\right) = E(tx)E(tx^p)\cdots E\left(tx^{p^{k-1}}\right). \end{aligned}$$

The lemma is proved.

Corollary 2.4 *If π is a root of $\sum_{i=0}^{\infty} t^{p^i}/p^i = 0$ in $\overline{\mathbb{Q}}_p$ with order $1/(p^{l-1}(p-1))$, and $x \in \overline{\mathbb{Q}}_p$ satisfies $x^{p^k} = x$, then*

$$E(\pi)^{x+x^p+\cdots+x^{p^{k-1}}} = E(\pi x)E(\pi x^p)\cdots E\left(\pi x^{p^{k-1}}\right).$$

We now fix an embedding of $\overline{\mathbb{Q}}$ into $\overline{\mathbb{Q}}_p$. Guaranteed by Lemma 2.2, we may choose, for each $l = 1, \ldots, m$, a unique root π_l of $\sum_{i=0}^{\infty} t^{p^i}/p^i = 0$ in $\overline{\mathbb{Q}}_p$ with order $1/(p^{l-1}(p-1))$

such that $E(\pi_l) = \psi(1)^{p^{m-l}}$. Let $\Delta = \Delta_\infty(f)$, $D = D(\Delta)$, and π a D-th root of $\pi_m^{p^{m-1}}$ in $\overline{\mathbb{Q}}_p$. For $b \geq 0$, we write

$$L(b) = \left\{ \sum_{u \in L(\Delta)} a_u x^u : a_u \in \mathbb{Z}_p[\mu_{q-1}, \pi_m, \pi],\ \mathrm{ord}_p(a_u) \geq b \deg(u) \right\}.$$

The Galois group $\mathrm{Gal}(\mathbb{Q}_p[\mu_{q-1}, \pi_m, \pi]/\mathbb{Q}_p)$ acts on $L(b)$ coefficientwise. Define

$$E_f(x) = \prod_{i=0}^{m-1} \prod_{u \in I_i} E(\pi_{m-i}\omega(a_{iu})x^u).$$

Lemma 2.5 *We have $E_f(x) \in L(1/(p-1))$.*

Proof Suppose that $0 \leq i \leq m-1$ and $u \in I_i$. We have $p^{m-i-1}u \in \Delta$. So $\deg(p^{m-i-1}u) \leq 1$, and

$$\mathrm{ord}_p(\pi_{m-i}) = \frac{1}{p^{m-i-1}(p-1)} \geq \frac{\deg(p^{m-i-1}u)}{p^{m-i-1}(p-1)} = \frac{\deg(u)}{p-1}.$$

It follows that $\pi_{m-i}\omega(a_{iu})x^u \in L(1/(p-1))$. Since $E(t) \in \mathbb{Z}_p[[t]]$, we have $E(\pi_{m-i}\omega(a_{iu})x^u) \in L(1/(p-1))$. The lemma now follows.

Let σ be the Frobenius element of $\mathrm{Gal}(\mathbb{Q}_p[\mu_{q-1}, \pi_m, \pi]/\mathbb{Q}_p)$ fixing π_m and π. The following lemma follows from Corollary 2.4.

Lemma 2.6 *If k is a positive integer, and $x \in \mu_{q^k-1}^n$, then*

$$\psi(Tr_{\mathbb{Q}_p[\mu_{q^k-1}]/\mathbb{Q}_p}(\omega(f)(x))) = \prod_{i=0}^{ak-1} E_f^{\sigma^i}(x^{p^i}).$$

Corollary 2.7 *We have*

$$S_k(f) = (-1)^{n-1} \sum_{x \in \mu_{q^k-1}^n} \prod_{i=0}^{ak-1} E_f^{\sigma^i}(x^{p^i}), \quad k = 1, 2, \ldots.$$

3 Functions from the Artin–Hasse exponential series

We shall study the growth of the coefficients of $\widehat{{}_kf}$ $(k = 1, \ldots, n)$, which are defined by

$$d \log \widehat{E}_f(x) = \sum_{k=1}^{n} \widehat{{}_kf} \frac{\mathrm{d}x_k}{x_k}, \quad \widehat{E}_f(x) = \prod_{j=0}^{\infty} E_f^{\sigma^j}(x^{p^j}).$$

Lemma 3.1 *We have*

$$\widehat{{}_kf} = \sum_{i=0}^{m-1} \sum_{j=0}^{\infty} p^j \gamma_{i,j} \sum_{u \in I_i} u_k \omega\left(a_{iu}^{p^j}\right) x^{p^j u}, \quad k = 1, \ldots, n,$$

where $\gamma_{i,j} = \sum_{l=0}^{j} \frac{\pi_{m-i}^{p^l}}{p^l}$.

Lemma 3.2 *We have* $\pi_l \equiv \pi_m^{p^{m-l}} \left(mod\ \pi_m^{p^{m-l}+1} \right)$.

Since $E(t) \in 1 + t + t^2\mathbb{Z}_p[[t]]$, we have $E(\pi_l) \equiv 1 + \pi_l (\bmod \pi_l^2)$. So we have

$$E(\pi_m)^{p^{m-l}} \equiv (1+\pi_m)^{p^{m-l}} \equiv 1 + \pi_m^{p^{m-l}} \left(\bmod \pi_m^{p^{m-l}+1} \right),$$

which, combined with the equality $E(\pi_l) = E(\pi_m)^{p^{m-l}}$, implies that $\pi_l \equiv \pi_m^{p^{m-l}} \left(\bmod \pi_m^{p^{m-l}+1} \right)$.

Corollary 3.3 *We have* $\pi_{m-i}^{p^j} \equiv \pi_m^{p^{i+j}} \left(mod\ \pi_m^{p^{i+j}+1} \right)$.

Lemma 3.4 *If* $j \le m-i-1$ *and* $l < j$, *we have*

$$ord_p \left(\frac{\pi_{m-i}^{p^l}}{p^l} \right) > ord_p \left(\frac{\pi_{m-i}^{p^j}}{p^j} \right).$$

Corollary 3.5 *If* $j \le m-i-1$, *we have* $p^j \gamma_{i,j} \equiv \pi_m^{p^{i+j}} \left(mod\ \pi_m^{p^{i+j}+1} \right)$.

Corollary 3.6 *Suppowse that* $j \le m-i-1$. *Then* $ord_p(p^j\gamma_{i,j}) > \frac{\deg(p^j u)}{p-1}$ *if* $\deg(p^{m-i-1}u) \le 1$, *and* $ord_p(p^j\gamma_{i,j} - \pi^{D\deg(p^j u)}) > \frac{\deg(p^j u)}{p-1}$ *if* $\deg(p^{m-i-1}u) = 1$.

Lemma 3.7 *If* $j \ge m-i$, *we have*

$$ord_p(p^j\gamma_{i,j}) - \frac{\deg(p^j u)}{p-1} \ge p^{j-(m-i)+1} - 1.$$

Proof Since $\gamma_{i,j} = -\sum_{l=j+1}^{\infty} \frac{\pi_{m-i}^{p^l}}{p^l}$, and $\mathrm{ord}_p(\frac{\pi_{m-i}^{p^l}}{p^l}) \ge \frac{p^{j+1}}{p^{m-i-1}(p-1)} - j + 1$ when $j \ge m-i$ and $l \ge j+1$, we have $\mathrm{ord}_p(p^j\gamma_{i,j}) \ge \frac{p^{j+1}}{p^{m-i-1}(p-1)} - 1$ if $j \ge m-i$. The lemma now follows from the fact that $\deg(p^{m-i-1}u) \le 1$.

Write

$$B = \left\{ \sum_{u \in L(\Delta)} a_u x^u \in L\left(\frac{1}{p-1}\right) : 0 \le \mathrm{ord}_p(a_u) - \frac{\deg(u)}{p-1} \to +\infty \text{ as } \deg(u) \to \infty \right\}.$$

Corollary 3.8 *For* $k = 1, \ldots, n$, *we have* $\widehat{{}_k f} \in B$, *and*

$$\widehat{{}_k f} \equiv \sum_{i=0}^{m-1} \sum_{j=0}^{m-i-1} \sum_{\deg(p^{m-i-1}u)=1} u_k \omega\left(a_{iu}^{p^j}\right) \pi^{D\deg(p^j u)} x^{p^j u} \ (mod\ \pi B).$$

4 The *p*-adic trace formula

We shall relate the L-function $L_f(t)$ to the characteristic polynomials of an operator $(p^n F^{-1})^a$ on p-adic spaces.

Since $E_f(x) \in L(1/(p-1))$ (Lemma 3.1), and $\psi_p : \sum_{u \in L(\Delta)} a_u x^u \mapsto \sum_{u \in L(\Delta)} a_{pu} x^u$ maps $L(b)$ to $L(pb)$, we have the following lemma.

Lemma 4.1 *The map $p^nF^{-1}: g \mapsto \sigma^{-1}\circ\psi_p(E_f(x)g)$ sends $L(1/(p-1))$ to $L(\frac{p}{p-1})$. In particular, p^nF^{-1} acts on B.*

Note that p^nF^{-1} is σ^{-1}-linear, and $(p^nF^{-1})^a = \psi_p^a \circ \prod_{i=0}^{a-1} E_f^{\sigma^i}(x^{p^i})$ is $\mathbb{Z}_p[\mu_{q-1}, \pi_m, \pi]$-linear. Write

$$\prod_{i=0}^{ak-1} E_f^{\sigma^i}(x^{p^i}) = \sum_{u\in L(\Delta)} a_u x^u.$$

Then the trace of $(p^nF^{-1})^{ak}$ on B is $\sum_{u\in L(\Delta)} a_{(q^k-1)u}$. And

$$S_k(f) = (-1)^{n-1}\left(q^k-1\right)^n \sum_{u\in L(\Delta)} a_{(q^k-1)u}.$$

So we have the following preliminary trace formula.

Proposition 4.2 *For $k = 1, 2, \ldots$, we have*

$$S_k(f) = -\left(1-q^k\right)^n Tr\left((p^nF^{-1})^{ak}; B\right).$$

Equivalently,

$$L_f(t) = \prod_{i=0}^{n} \det\left(1-(p^nF^{-1})^a q^i t; B\right)^{(-1)^i\binom{n}{i}}$$

Let $e_1 = (1,0,\ldots,0), \ldots, e_n = (0,\ldots,0,1)$. For $l = 0, 1, \ldots, n$, we write

$$K_l = \bigoplus_{1\le i_1<\cdots<i_l\le n} Be_{i_1}\wedge\cdots\wedge e_{i_l}$$

and define

$$p^nF^{-1}: K_l \to K_l, \quad ge_{i_1}\wedge\cdots\wedge e_{i_l} \mapsto p^{l+n}F^{-1}(g)e_{i_1}\wedge\cdots\wedge e_{i_l}.$$

Then the preliminary trace formula takes the following form.

Proposition 4.3 *For $k = 1, 2, \ldots$, we have*

$$S_k(f) = \sum_{l=0}^{n}(-1)^{l+1} Tr\big((p^nF^{-1})^{ak}; K_l\big).$$

By Corollary 3.8, $\hat{D}_j : g \mapsto (x_j\frac{\partial}{\partial x_j} + \widehat{jf})g$, $j = 1, \ldots, n$, operate on B. Obviously, they commute with each other. So, for $l = 1, \ldots, n$,

$$\hat{\partial}: K_l \to K_{l-1}, \quad ge_{i_1}\wedge\cdots\wedge e_{i_l} \mapsto \sum_{k=1}^{l}(-1)^{k-1}\hat{D}_{i_k}(g)e_{i_1}\wedge\cdots\wedge\hat{e}_{i_k}\wedge\cdots\wedge e_{i_l},$$
$$i_1 < \cdots < i_l$$

are well-defined, and satisfy $\hat{\partial}^2 = 0$. Thus we get a complex

$$K_n \xrightarrow{\hat{\partial}} K_{n-1} \xrightarrow{\hat{\partial}} \cdots \xrightarrow{\hat{\partial}} K_0.$$

It is easy to check that $p^nF^{-1}\circ\hat{\partial} = \hat{\partial}\circ p^nF^{-1}$. That is, p^nF^{-1} operates on the complex $(K_\bullet, \hat{\partial})$. Therefore we have the following homological trace formula.

Proposition 4.4 *For $k = 1, 2, \ldots$, we have*

$$S_k(f) = \sum_{l=0}^{n} (-1)^{l+1} Tr\left((p^n F^{-1})^{ak}; H_l(K_\bullet, \hat{\partial})\right).$$

Equivalently,

$$L_f(t) = \prod_{l=0}^{n} \det\left(1 - (p^n F^{-1})^a t; H_l(K_\bullet, \hat{\partial})\right)^{(-1)^l}.$$

5 The total degree of the L-function

We shall study the Newton polygon of $\det(1 - (p^n F^{-1})^a t; B)$, and then prove Theorem 1.2.

Proposition 5.1 *The Newton polygon of* $\det(1 - (p^n F^{-1})^a t; B)$ *with respect to* ord_q *lies above the Hodge polygon of* $\sum_{i=0}^{+\infty} W_\Delta(k) t^k$ *of degree* D.

Write $E_f(x) = \sum_{u \in L(\Delta)} a_u \pi^{D \deg(u)} x^u$, $a_u \in \mathbb{Z}_p[\mu_{q-1}, \pi_m, \pi]$. Then the matrix of $p^n F^{-1}$ with respect to the orthonormal basis $\{\pi^{D \deg(u)} x^u\}_{u \in L(\Delta)}$, written as a column vector, is

$$A^{\sigma^{-1}} = \left(a_{pw-u}^{\sigma^{-1}} \pi^{D((p-1)\deg(w) + c(w,u))}\right)_{w,u}, \quad c(w,u) = \deg(pw-u) + \deg(u) - p\deg(w) \geq 0.$$

So, the matrix of $(p^n F^{-1})^a$ with respect to that orthonormal basis is $AA^\sigma \ldots A^{\sigma^{a-1}}$. Obviously, the Newton polygon of $\det(1 - At)$ with respect to ord_p lies above the polygon with vertices at points $(0, 0)$ and

$$\left(\sum_{i=0}^{k} W_\Delta(i), \sum_{i=0}^{k} W_\Delta(i) \frac{i}{D}\right), \quad k = 0, 1, \ldots.$$

It follows that the Newton polygon of $\det(1 - (p^n F^{-1})^a t; B) = \det(1 - AA^\sigma \cdots A^{\sigma^{a-1}} t)$ with respect to ord_q lies above the polygon with vertices at points $(0, 0)$ and

$$\left(\sum_{i=0}^{k} W_\Delta(i), \sum_{i=0}^{k} W_\Delta(i) \frac{i}{D}\right), \quad k = 0, 1, \ldots.$$

The proposition is proved.

Corollary 5.2 *If $j \leq n + 1$, then* $\det(1 - (p^n F^{-1})^a t; B)$ *has at most* $\sum_{k=0}^{Dj} W_\Delta(k)$ *zeros of q-order $\leq j - 1$.*

Proof Define

$$\sum_{k=0}^{+\infty} h_\Delta(k) t^k = (1-t)^n \sum_{k=0}^{+\infty} W_\Delta(k) t^k.$$

Since $\sum_{k=0}^{+\infty} h_\Delta(k)t^k$ is a polynomial of degree $\le n$ with non-negative coefficients by a lemma of Kouchnirenko [10, Lemma 2.9], and

$$\sum_{k=0}^{jD-i}\binom{n-1+k}{n-1}(k+i)=\binom{n+Dj-i}{n}\left(\frac{n(Dj-i)}{n+1}+i\right)\ge\binom{n+Dj-i}{n}D(j-1),$$

we have

$$\begin{aligned}\frac{1}{D}\sum_{k=0}^{jD}kW_\Delta(k)&=\frac{1}{D}\sum_{k=0}^{jD}k\sum_{i=0}^{k}h_\Delta(i)\binom{n-1+k-i}{n-1}\\&=\frac{1}{D}\sum_{i=0}^{n}h_\Delta(i)\sum_{k=i}^{jD}\binom{n-1+k-i}{n-1}k\\&=\frac{1}{D}\sum_{i=0}^{n}h_\Delta(i)\sum_{k=0}^{jD-i}\binom{n-1+k}{n-1}(k+i)\\&\ge(j-1)\sum_{i=0}^{n}h_\Delta(i)\sum_{k=0}^{jD-i}\binom{n-1+k}{n-1}\ge(j-1)\sum_{k=0}^{jD}W_\Delta(k).\end{aligned}$$

The corollary now follows from the above inequality by Proposition 5.1.

We now prove Theorem 1.2. Since the reciprocal zeros and reciprocal poles of $L_f(t)$ are of q-order $\le n$, its total number, by the preliminary trace formula, is bounded by the number of reciprocal zeros of $\prod_{i=0}^{n}\det(1-(p^nF^{-1})^aq^it;B)^{\binom{n}{i}}$. By Corollary 5.2, that number is bounded by

$$\sum_{i=0}^{n}\binom{n}{i}\sum_{k=0}^{D(n-i+1)}W_\Delta(k).$$

Theorem 1.2 is proved.

6 The acyclicity of the *p*-adic complex

In this section we shall prove the following proposition, which implies the first statement of Theorem 1.3.

Proposition 6.1 *If f non-degenerate with respect to $\Delta_\infty(f)$, then $(K_\bullet,\hat\partial)$ is acyclic at positive dimensions, and $H_0(K_\bullet,\hat\partial)$ is a $\mathbb{Z}_p[\mu_{q-1},\pi_m,\pi]$-module free of rank $n!Vol(\Delta_\infty(f))$.*

Write

$$\bar B:=\mathbb{F}_q[x^{L(\Delta)}]:=\left\{\sum_{u\in L(\Delta)}a_ux^u:a_u\in\mathbb{F}_q\right\}.$$

It is a ring with the multiplication rule

$$x^ux^{u'}=\begin{cases}x^{u+u'}, & \text{if } u \text{ and } u' \text{ are cofacial,}\\ 0, & \text{otherwise.}\end{cases}$$

Define

$$B \to \bar{B},\ \sum_{u\in L(\Delta)} a_u \pi^{D\deg(u)} x^u \mapsto \sum_{u\in L(\Delta)} \bar{a}_u x^u,$$

where $\bar{a}_u$ is the residue class of a_u modulo the maximal ideal of $\mathbb{Z}_p[\mu_{q-1}, \pi_m, \pi]$.

Lemma 6.2 *The map $B \to \bar{B}$ is a ring homomorphism. And the sequence*

$$0 \to B \to B \to \bar{B} \to 0$$

is exact.

For $j = 1, \ldots, n$, we define

$$\bar{D}_j : \bar{B} \to \bar{B},\ g \mapsto \left(x_j \frac{\partial}{\partial x_j} + \overline{{}_j f} \right) g,$$

where

$$\overline{{}_j f} = \sum_{i=0}^{m-1} \sum_{j=0}^{m-i-1} \sum_{\deg(p^{m-i-1}u)=1} u_k a_{iu}^{p^j} x^{p^j u}.$$

By Corollary 3.8, we have the following lemma.

Lemma 6.3 *For $j = 1, \ldots, n$, the diagram*

$$\begin{array}{ccc} B & \to & \bar{B} \\ \hat{D}_j \downarrow & & \bar{D}_j \downarrow \\ B & \to & \bar{B} \end{array}$$

is commutative.

For $l = 0, \ldots, n$, we define

$$\bar{K}_l = \bigoplus_{1\le i_1<\cdots<i_l\le n} \bar{B} e_{i_1} \wedge \cdots \wedge e_{i_l}.$$

For $l = 1, \ldots, n$, we define

$$\bar{\partial} : \bar{K}_l \to \bar{K}_{l-1}, \quad g e_{i_1} \wedge \cdots \wedge e_{i_l} \mapsto \sum_{k=1}^{l} (-1)^{k-1} \bar{D}_{i_k}(g) e_{i_1} \wedge \cdots \wedge \hat{e}_{i_k} \wedge \cdots \wedge e_{i_l},$$
$$i_1 < \cdots < i_l.$$

It is easy to see that the sequence

$$\bar{K}_n \xrightarrow{\bar{\partial}} \bar{K}_{n-1} \xrightarrow{\bar{\partial}} \cdots \xrightarrow{\bar{\partial}} \bar{K}_0$$

is a complex.

Proposition 6.4 *The map $B \to \bar{B}$ induces a morphism of complexes from $(K_\bullet, \hat{\partial})$ to $(\bar{K}_\bullet, \bar{\partial})$. Moreover, the sequence*

$$0 \to (K_\bullet, \hat{\partial}) \to (K_\bullet, \hat{\partial}) \to (\bar{K}_\bullet, \bar{\partial}) \to 0$$

is exact.

Proof The first statement follows from Lemma 6.3, and the second follows from Lemma 6.2.

By Proposition 6.4, and a lemma of Monsky [14, Theorem 8.5], the proof of Proposition 6.1 is reduced to the proof of the following proposition.

Proposition 6.5 *If f is non-degenerate with respect to $\Delta_\infty(f)$, then $(\bar{K}_\bullet, \bar{\partial})$ is acyclic at positive dimensions, and $H_0((\bar{K}_\bullet, \bar{\partial}))$ is a $\mathbb{F}_q$-vector space of dimension $n!Vol(\Delta_\infty(f))$.*

For $j = 1, \ldots, n$, we define

$$\overline{{}_j f}^0 = \sum_{i=0}^{m-1} \sum_{\deg(p^{m-i-1}u)=1} u_k a_{iu}^{p^{m-i-1}} x^{p^{m-i-1}u}.$$

For $l = 1, \ldots, n$, we define

$$\bar{\partial}^0 : \bar{K}_l \to \bar{K}_{l-1}, \quad g e_{i_1} \wedge \cdots \wedge e_{i_l} \mapsto \sum_{k=1}^{l} (-1)^{k-1} \overline{{}_{i_k} f}^0 g e_{i_1} \wedge \cdots \wedge \hat{e}_{i_k} \wedge \cdots \wedge e_{i_l},$$
$$i_1 < \cdots < i_l.$$

Then

$$\bar{K}_n \xrightarrow{\bar{\partial}^0} \bar{K}_{n-1} \xrightarrow{\bar{\partial}^0} \cdots \xrightarrow{\bar{\partial}^0} \bar{K}_0$$

is a complex. In the next section, we shall prove the following proposition.

Proposition 6.6 *If f is non-degenerate with respect to $\Delta := \Delta_\infty(f)$, then the complex $(\bar{K}_\bullet, \bar{\partial}^0)$ is acyclic at positive dimensions, and the Poincaré series of $H_0((\bar{K}_\bullet, \bar{\partial}^0))$ is $P_\Delta(t)$. In particular, $H_0((\bar{K}_\bullet, \bar{\partial}^0))$ is a $\mathbb{F}_q$-vector space of dimension $n!Vol(\Delta)$.*

We now deduce the first statement of Proposition 6.5 from Proposition 6.6. In a given a homology class of positive dimension, we choose one representative ξ of lowest degree. We claim that $\xi = 0$. Otherwise, let ξ^0 be the leading term of ξ. We have $\bar{\partial}^0(\xi^0) = 0$ since it is the leading term of $\bar{\partial}(\xi) = 0$. By the acyclicity of $(\bar{K}_\bullet, \bar{\partial}^0)$, $\xi^0 = \bar{\partial}^0(\eta)$ for some η. The form $\xi - \bar{\partial}(\eta)$ is now of lower degree than ξ, contradicting to our choice of ξ. The proposition is proved.

The second statement of Proposition 6.5 follows the following proposition.

Proposition 6.7 *Let V be a basis of $\bar{K}_0$ modulo $\bar{\partial}^0(\bar{K}_1)$ consisting of homogeneous elements. Then V is also a basis of $\bar{K}_0$ modulo $\bar{\partial}(\bar{K}_1)$.*

Proof First, we show that $\bar{K}_0$ is generated by V and $\bar{\partial}(\bar{K}_1)$. Otherwise, among elements of $\bar{K}_0$ which are not linear combinations of elements of V and $\bar{\partial}(\bar{K}_1)$, we choose one of lowest degree. We may suppose that it is of form $\bar{\partial}^0(\xi)$. Let ξ^0 be the leading term of ξ. Then $\bar{\partial}^0(\xi) - \bar{\partial}(\xi^0)$ is not a linear combination of elements of V and $\bar{\partial}(\bar{K}_1)$, and is of lower degree than $\partial^0(\xi)$. This is a contradiction. Therefore $\bar{K}_0$ is generated

by E and $\bar{\partial}(\bar{K}_1)$. It remains to show that $\xi = 0$ whenever ξ belongs to $\bar{\partial}(\bar{K}_1)$ and is a linear combination of elements of V. Otherwise, we may choose one element ζ of lowest degree such that $\xi = \bar{\partial}(\zeta)$. Let ζ^0 be the leading term of ζ. Then $\bar{\partial}^0(\zeta^0)$ is a linear combination of elements of V since it is the leading term of $\bar{\partial}(\zeta)$. So we have $\bar{\partial}^0(\zeta^0) = 0$. By the acyclicity of $(\bar{K}_\bullet, \bar{\partial}^0)$, $\zeta^0 = \bar{\partial}^0(\eta)$ for some η. The form $\zeta - \bar{\partial}(\eta)$ is now of lower degree than ζ, contradicting to our choice of ζ. This completes the proof of the proposition.

7 The complex obtained by reduction

In this section, we shall prove Proposition 6.6. The second statement follows from the first, and the last follows from the second and a lemma of Kouchnirenko [10, Lemma 2.9]. So it remains to prove the acyclicity of the complex $(\bar{K}_\bullet, \bar{\partial}^0)$.

Let τ be a face of Δ that does not contain the origin, and $\bar{\tau}$ is the convex hull in $\mathbb{Q}^n$ generated by τ and the origin. For $\alpha_1, \ldots, \alpha_s$ in $\mathbb{F}_q[x^{L(\bar{\tau})}]$, we define $\bar{K}_\bullet(\bar{\tau}, \{\alpha_j\}_{j=1}^s)$ to be the complex

$$\bar{K}_l(\bar{\tau}, \{\alpha_j\}_{j=1}^s) = \bigoplus_{1 \le i_1 < \cdots < i_l \le s} \mathbb{F}_q[x^{L(\bar{\tau})}] e_{i_1} \wedge \cdots \wedge e_{i_l}, \quad l = 0, \ldots, s$$

with derivation

$$g e_{i_1} \wedge \cdots \wedge e_{i_l} \mapsto \sum_{k=1}^{l} (-1)^{k-1} \alpha_{i_k} g e_{i_1} \wedge \cdots \wedge \hat{e}_{i_k} \wedge \cdots \wedge e_{i_l},\ 1 \le i_1 < \cdots < i_l \le s.$$

By a proposition of Kouchnirenko [10, Proposition 2.6] and the argument of Adolphson–Sperber [2, p.379], the sequence

$$0 \to \bar{K}_\bullet^{\emptyset}(f) \to \bigoplus_{\dim \tau = n-1} \bar{K}_\bullet\left(\bar{\tau}, \{\overline{{}_j f^\tau}\}_{j=1}^n\right) \to \cdots \to \bigoplus_{\dim \tau = 0} \bar{K}_\bullet\left(\bar{\tau}, \{\overline{{}_j f^\tau}\}_{j=1}^n\right) \to \bar{K}_\bullet^{-1} \to 0$$

is exact, where τ denotes a face of Δ which is not contained in any $(n-1)$-dimensional face that does not contain the origin, and

$$\bar{K}_l^{-1} = \begin{cases} \mathbb{F}_q^{\binom{n}{l}}, & \text{if the origin is in the interior of } \Delta \text{ and } 1 \le l \le n, \\ 0, & \text{otherwise.} \end{cases}$$

By the exactness of that sequence, the acyclicity of the complex $(\bar{K}_\bullet, \bar{\partial}^0)$ follows from the following lemma.

Lemma 7.1 *Let f be a Witt vector of length m with coefficients in $\mathbb{F}_q[x_1^{\pm 1}, \ldots, x_n^{\pm 1}]$. Suppose that f is non-degenerate with respect to $\Delta := \Delta_\infty(f)$ and $\dim \Delta = n$. Let τ be a face of Δ of dimension $s-1$ that does not contain the origin. Then the complex $\bar{K}_\bullet(\bar{\tau}, \{\overline{{}_j f^\tau}\}_{j=1}^n)$ is acyclic at dimensions $> n - s$.*

Since the sequence

$$0 \to \bar{K}_\bullet\left(\bar{\tau}, \{\alpha_j\}_{j=1}^{s-1}\right) \to \bar{K}_\bullet\left(\bar{\tau}, \{\alpha_j\}_{j=1}^{s}\right) \to \bar{K}_\bullet\left(\bar{\tau}, \{\alpha_j\}_{j=1}^{s-1}\right)[-1] \to 0$$

is exact, Lemma 7.1 follows from the following one.

Lemma 7.2 *Suppose that f is non-degenerate with respect to $\Delta_\infty(f)$. If τ is a face of Δ of dimension $r-1$ that does not contain the origin, then there are $1 \le i_1 < \cdots < i_r \le n$ such that the complex $\bar{K}_\bullet(\bar{\tau}, \{\overline{{}_{ij}f}^{\tau}\}_{j=1}^{r})$ is acyclic at positive dimensions.*

Proof There are $\{i_1, \ldots, i_r\} \subset \{1, \ldots, n\}$ and $(\alpha_{kj}) \in \mathbb{Q} \cap \mathbb{Z}_p$ $(1 \le k \le n, 1 \le j \le r)$ such that $u_k = \alpha_{1k}u_{i_1} + \cdots + \alpha_{rk}u_{i_r}$ for all $u = (u_1, \ldots, u_n) \in L(\bar{\tau})$. Let σ be any face of τ. We have

$$\overline{{}_{j}f}^{\sigma} = \alpha_{1ji_1}\overline{{}_{i_1}f}^{\sigma} + \cdots + \alpha_{rji_r}\overline{{}_{i_r}f}^{\sigma}.$$

So $\overline{{}_{i_1}f}^{\sigma}, \ldots, \overline{{}_{i_r}f}^{\sigma}$ have no common zeros in $(\overline{\mathbb{F}}_q^{\times})^n$. By a theorem of Kouchnirenko [10, Theorem 6.2], $\overline{{}_{i_1}f}^{\tau}, \ldots, \overline{{}_{i_r}f}^{\tau}$ generate in $\mathbb{F}_q[x^{L(\bar{\tau})}]$ an ideal of finite codimension. Note that $\mathbb{F}_q[x^{L(\bar{\tau})}]$ is Cohen–Macaulay by a theorem of Hochster [8, Theorem 1]. The complex $\bar{K}_\bullet(\bar{\tau}, \{\overline{{}_{ij}f}^{\tau}\}_{j=1}^{r})$ is acyclic at positive dimensions by a theorem of Serre [15, Theorem 3, Chap. IV]. The lemma is proved.

8 The Newton polygon of the *L*-function

In this section we shall prove the second statement of Theorem 1.3. (The last statement follows from the second by a lemma of Kouchnirenko [10, Lemma 2.9].) By the argument of Dwork [7, §7], it suffices to prove the following proposition.

Proposition 8.1 *If f is non-degenerate with respect to $\Delta := \Delta_\infty(f)$, then the Newton polygon of* $\det(1-p^nF^{-1}t; H_0(K_\bullet, \hat{\partial}))$ *with respect to ord_p lies above the Hodge polygon of $P_\Delta(t)$ of degree $D(\Delta)$, and their endpoints coincide.*

Let $\bar{V}$ be a basis of $\bar{K}_0$ modulo $\bar{\partial}^0(K_1)$ consisting of homogeneous elements. By Proposition 6.7, it is also a basis of $\bar{K}_0$ modulo $\bar{\partial}(K_1)$. Define

$$V = \left\{ \sum \omega(a_u)x^u : \sum a_u x^u \in \bar{V} \right\}.$$

It is a basis of B modulo $\sum_{k=1}^n \hat{D}_k B$. For real numbers $b > 1/(p-1)$ and c, we write

$$L(b,c) = \left\{ \sum_{u \in L(\Delta)} a_u x^u : a_u \in \mathbb{Q}_p[\mu_{q-1}, \pi_m, \pi],\ \mathrm{ord}_p(a_u) \ge b\deg(u) + c \right\}.$$

It is compact with respect to the topology of coefficientwise convergence. Let $V(b,c)$ be the subset of elements of $L(b,c)$ which are finite linear combinations of elements of V. In the next section we shall prove the following proposition.

Proposition 8.2 *If $1/(p-1) < b < p/(p-1)$, then*

$$L(b,c) = V(b,c) + \sum_{k=1}^{n} \hat{D}_k L\left(b, c+b-\frac{1}{p-1}\right).$$

We now prove the first statement of Proposition 8.1. For each $\xi \in V$, we write

$$p^nF^{-1}\left(\pi^{D\deg(\xi)}\xi\right) \equiv \sum_{\eta \in V} c_{\eta,\xi}\pi^{D\deg(\eta)}\eta \left(\mathrm{mod} \sum_{k=1}^{n} \hat{D}_k B\right),\ c_{\eta,\xi} \in \mathbb{Z}_p[\mu_{q-1}, \pi_m, \pi]$$

By Lemma 4.1, $p^nF^{-1}(\pi^{D\deg(\xi)}\xi)$ lies in the space $L(\frac{p}{p-1})$. So, by Proposition 8.2, $c_{\eta,\xi}\pi^{D\deg(\eta)}\eta$ lies in every $L(b)$ with $1/(p-1) < b < \frac{p}{p-1}$. That is, $\mathrm{ord}_p(c_{\eta,\xi}) \geq (b-1/(p-1))\deg(\eta)$ for every $1/(p-1) < b < \frac{p}{p-1}$. Thus we have $\mathrm{ord}_p(c_{\eta,\xi}) \geq \deg(\eta)$. Therefore, the Newton polygon of the characteristic polynomial of $(c_{\eta,\xi})$, which is now the Newton polygon of $\det(1-p^nF^{-1}t; H_0(K_\bullet, \hat{\partial}))$, lies above the Hodge polygon of $P_\Delta(t)$ of degree D. In particular, $\mathrm{ord}_p(\det(c_{\eta,\xi})) \geq \sum_{\xi\in V}\deg(\xi)$.

It remains to show that the Newton polygon of $\det(1-p^nF^{-1}t; H_0(K_\bullet, \hat{\partial}))$ share the same endpoints with the Hodge polygon of $P_\Delta(t)$ of degree D. Define

$$\phi_p : L\left(\frac{p}{p-1}\right) \to L\left(\frac{1}{p-1}\right), \quad \sum_{u\in L(\Delta)} a_u x^u \mapsto \sum_{u\in L(\Delta)} a_u x^{pu}.$$

Obviuosly, $p^nF^{-1}\circ(E_f^{-1}\circ\phi_p\circ\sigma) = 1$ on $L(\frac{p}{p-1})$. At the end of this we shall prove the following proposition.

Proposition 8.3 *Modulo $\sum_{k=1}^n \hat{D}_k L(\frac{1}{p-1})$, the space $L(\frac{1}{p-1})$ is generated by $\{\pi^{D\deg(\xi)}\xi : \xi \in V\}$.*

So, for each $\xi \in V$, we can find $b_{\eta,\xi} \in \mathbb{Z}_p[\mu_{q-1}, \pi_m, \pi]$

$$E_f^{-1}\circ\phi_p\circ\sigma\left(\pi^{D\deg(\xi)p}\xi\right) \equiv \sum_{\eta\in V} b_{\eta,\xi}\pi^{D\deg(\eta)}\eta \left(\mathrm{mod}\sum_{k=1}^n \hat{D}_k L\left(\frac{1}{p-1}\right)\right).$$

It follows that $(c_{\eta,\xi})(b_{\eta,\xi}) = \mathrm{diag}\{\pi^{D\deg(\xi)(p-1)}, \xi \in V\}$. So $\mathrm{ord}_p(\det(c_{\eta,\xi})) \leq \sum_{\eta\in V}\deg(\eta)$. Therefore

$$\mathrm{ord}_p(\det(c_{\eta,\xi})) = \sum_{\eta\in V}\deg(\eta).$$

That is, the Newton polygon of $\det(1-p^nF^{-1}t; H_0(K_\bullet, \hat{\partial}))$ share the same endpoints with the Hodge polygon of $P_\Delta(t)$ of degree D.

We now prove Proposition 8.3. Let $\xi = \sum_{u\in L(\Delta)} a_u x^u \in L(\frac{p}{p-1})$. For $N = 0, 1, \ldots$, write $\xi^{(N)} = \sum_{u\in L(\Delta),\deg(u)\leq N} a_u x^u \in B$. As $\{\pi^{D\deg(\eta)}\eta : \eta \in V\}$ is a basis of B modulo $\sum_{k=1}^n \hat{D}_k B$, there are elements $\xi_k^{(N)} \in B$ $(k = 1, \ldots, n)$ such that

$$\xi^{(N)} - \sum_{k=1}^n \hat{D}_k\xi_k^{(N)} = \sum_{\eta\in V} a_\eta^{(N)}\pi^{D\deg(\eta)}\eta.$$

As $L(1/(p-1))$ is compact with respect to the topology of coefficientwise convergence, the sequence $(\{\xi_k^{(N)}\}_{k=1}^n, \{a_\eta^{(N)}\}_{\eta\in V})$, $N = 0, 1, \ldots$, has an adherent point $(\{\xi_k\}_{k=1}^n, \{a_\eta\}_{\eta\in V})$ in the space $L(b)^n \times (\mathbb{Z}_p[\mu_{q-1}, \pi_m, \pi])^{|V|}$. Therefore we get

$$\xi - \sum_{k=1}^n \hat{D}_k\xi_k = \sum_{\eta\in V} a_\eta\pi^{D\deg(\eta)p^{m-1}}\eta.$$

This completes the proof of Proposition 8.3.

9 The space $L(b,c)$

In this section we shall prove Propositions 8.2.

For $k=1,\dots,n$, we write

$$\widehat{{}_kf}^0=\sum_{i=0}^{m-1}p^{m-i-1}\gamma_{i,m-i-1}\sum_{\deg(p^{m-i-1}u)=1}u_k\omega\left(a_{iu}^{p^{m-i-1}}\right)x^{p^{m-i-1}u}.$$

For $l=1,\dots,n$, we define

$$\hat{\partial}^0:K_l\to K_{l-1},\quad ge_{i_1}\wedge\cdots\wedge e_{i_l}\mapsto\sum_{k=1}^{l}(-1)^{k-1}\widehat{{}_{i_k}f}^0ge_{i_1}\wedge\cdots\wedge\hat{e}_{i_k}\wedge\cdots\wedge e_{i_l},$$
$$i_1<\cdots<i_l.$$

It is easy to see that

$$0\to(K_\bullet,\hat{\partial}^0)\to(K_\bullet,\hat{\partial}^0)\to(\bar{K}_\bullet,\bar{\partial})\to 0$$

is an exact sequence of complex. So we have the following lemma.

Lemma 9.1 *Modulo $\sum_{k=1}^n\widehat{{}_kf}^0B$, the space B is generated by $\{\pi^{D\deg(\xi)}\xi:\xi\in V\}$.*

Corollary 9.2 *If $b>1/(p-1)$, then*

$$L(b,c)=V(b,c)+\sum_{k=1}^{n}\widehat{{}_kf}^0L\left(b,c+b-\frac{1}{p-1}\right).$$

Proof Let $\xi\in L(b,c)$, ξ_v ($v\in\deg L(\Delta)$) its homogeneous part of degree v, and k_v the least integer such that $\mathrm{ord}_p(\pi^{k_v})\geq bv+c$. Then $\pi^{Dv-k_v}\xi_v\in B$. By the above lemma, we may write

$$\pi^{Dv-k_v}\xi_v=\sum_{\eta\in V,\deg(\eta)\leq v}a_\eta^{(v)}\pi^{D\deg(\eta)}\eta+\sum_{i=1}^{n}\widehat{{}_if}^0\eta_i^{(v)},$$

where $a_\eta^{(v)}\in\mathbb{Z}_p[\mu_{q-1},\pi_m,\pi]$, and $\eta_i^{(v)}\in B$ is of degree $\leq v-1$. It follows that

$$\xi=\sum_{\eta\in V}\eta\pi^{D\deg(\eta)}\sum_{v\geq\deg(\eta)}a_\eta^{(v)}\pi^{k_v-Dv}+\sum_{i=1}^{n}\widehat{{}_if}^0\sum_{v\in\deg L(\Delta)}\pi^{k_v-Dv}\eta_i^{(v)}.$$

It is easy to see that the first term on the right-hand side converges to an element in $V(b,c)$, and the inner sum in the second term converges to an element in $L(b,c+b-1/(p-1))$. The corollary is proved.

For $k=1,\dots,n$, we define

$$D_k:B\to B,\ g\mapsto\left(x_k\frac{\partial}{\partial x_k}+\sum_{i=0}^{m-1}\sum_{j=0}^{m-i-1}p^j\gamma_{i,j}\sum_{u\in I_i}u_k\omega(a_{iu}^{p^j})x^{p^ju}\right)g.$$

For $l=1,\dots,n$, we define

$$\partial:K_l\to K_{l-1},\quad ge_{i_1}\wedge\cdots\wedge e_{i_l}\mapsto\sum_{k=1}^{l}(-1)^{k-1}D_k(g)e_{i_1}\wedge\cdots\wedge\hat{e}_{i_k}\wedge\cdots\wedge e_{i_l},$$
$$i_1<\cdots<i_l.$$

Corollary 9.3 *If $b > 1/(p-1)$, then*

$$L(b,c) = V(b,c) + \sum_{k=1}^{n} D_k L(b, c+b-\frac{1}{p-1}).$$

Proof Note that $\widehat{{}_k f}^0 - D_k$ maps $L(b,c)$ to $L(b, c-(b-1/(p-1))e)$ for some constant $e<1$. Let $\xi \in L(b,c)$. By the previous corollary and induction, we can find a sequence

$$\left(\eta_0^{(i)}, \ldots, \eta_n^{(i)}\right) \in V\left(b, c+i(1-e)\left(b-\frac{1}{p-1}\right)\right) \times L\left(b, c+(i(1-e)+1)\left(b-\frac{1}{p-1}\right)\right)^n, \quad i=0,1,\ldots$$

such that

$$\xi = \eta_0^{(0)} + \sum_{k=1}^{n} \widehat{{}_k f}^0 \eta_k^{(0)},$$

and

$$\sum_{k=1}^{n} \left(\widehat{{}_k f}^0 - D_k\right) \eta_k^{(i)} = \eta_0^{(i+1)} + \sum_{k=1}^{n} \widehat{{}_k f}^0 \eta_k^{(i+1)}.$$

One sees immediately that $\sum_{i=0}^{\infty} \eta_0^{(i)}$ converges to an element η_0 in $V(b,c)$, and $\sum_{i=0}^{\infty} \eta_k^{(i)}$ converges to an element η_k in $L(b, c+b-1/(p-1))$. Moreover, we have $\xi = \eta_0 + \sum_{k=1}^{n} D_k \eta_k$. The corollary is proved.

We now prove Proposition 8.2. Note that $D_k - \hat{D}_k \in L(b, p(\frac{p}{p-1}-b)-1)$ by Lemma 3.7. So it maps $L(b,c)$ to the space $L(b, c+p(\frac{p}{p-1}-b)-1)$. Let $\xi \in L(b,c)$. By the previous corollary and induction, we can find a sequence

$$\left(\eta_0^{(i)}, \ldots, \eta_n^{(i)}\right) \in V(b, c+i(p-(p-1)b)) \times L\left(b, c+i(p-(p-1)b)+\left(b-\frac{1}{p-1}\right)\right)^n, \; i=0,1,\ldots$$

such that

$$\xi = \eta_0^{(0)} + \sum_{k=1}^{n} D_k \eta_k^{(0)},$$

and

$$\sum_{k=1}^{n} (D_k - \hat{D}_k) \eta_k^{(i)} = \eta_0^{(i+1)} + \sum_{k=1}^{n} D_k \eta_k^{(i+1)}.$$

One sees immediately that $\sum_{i=0}^{\infty} \eta_0^{(i)}$ converges to an element η_0 in $V(b,c)$, and $\sum_{i=0}^{\infty} \eta_k^{(i)}$ converges to an element η_k in $L(b, c+b-\frac{1}{p-1})$. Moreover, we have $\xi = \eta_0 + \sum_{k=1}^{n} D_k \eta_k$. This completes the proof of Proposition 8.2.

10 The weights of the *L*-function

We shall prove Theorem 1.4.

Let Δ be a convex polyhedron in $\mathbb{Q}^n$ of dimension n that contains the origin, and S_Δ the sum of the volumes of all its $(n-1)$-dimensional faces that contain 0. Write

$$(1-t^{D(\Delta)})^n \sum_{i=0}^{+\infty} W_\Delta(i)t^i = \sum_{i=0}^{D(\Delta)n} h_\Delta(i)t^i.$$

Lemma 10.1 *We have*

$$\frac{1}{D(\Delta)} \sum_{i=0}^{nD(\Delta)} ih_\Delta(i) = \frac{n}{2}n!Vol(\Delta) - \frac{(n-1)!}{2}S_\Delta.$$

In particular, $\frac{1}{D(\Delta)} \sum_{i=0}^{nD(\Delta)} ih_\Delta(i) = \frac{n}{2}n!Vol(\Delta)$ *if the origin is an interior point of* Δ.

Proof Note that

$$W_\Delta(i) = \sum_{k=0}^{i/D(\Delta)} h_\Delta(i - D(\Delta)k) \binom{n-1+k}{n-1}.$$

So

$$\begin{aligned}\sum_{i\le D(\Delta)x} \left(x - \frac{i}{D(\Delta)}\right) W_\Delta(i) &= \sum_{j=0}^{nD(\Delta)} h_\Delta(j) \sum_{0\le k\le x-\frac{j}{D(\Delta)}} \left(x - \frac{j}{D(\Delta)} - k\right)\binom{n-1+k}{n-1} \\ &= \sum_{j=0}^{nD(\Delta)} h_\Delta(j)\left(\frac{x^{n+1}}{(n+1)!} + \left(\frac{n}{2} - \frac{j}{D(\Delta)}\right)\frac{x^n}{n!} + O(x^{n-1})\right).\end{aligned}$$

On the other hand, by [1, (4.12–13)],

$$\sum_{i\le D(\Delta)x} \left(x - \frac{i}{D(\Delta)}\right) W_\Delta(i) = n!\mathrm{Vol}(\Delta)\frac{x^{n+1}}{(n+1)!} + \frac{(n-1)!}{2}S_\Delta\frac{x^n}{n!} + O(x^{n-1}).$$

The lemma now follows.

We now prove Theorem 1.4. Let α_i, $i = 1,\dots,n!\mathrm{Vol}(\Delta)$, be the eigenvalues of q^nF^{-1} on $H_0(K_\bullet, \hat{\partial})$. By Theorem 1.2 and Lemma 10.1,

$$\mathrm{ord}_q\left(\prod_{i=1}^{n!\mathrm{Vol}(\Delta)} \alpha_i\right) = \frac{n}{2}n!\mathrm{Vol}(\Delta).$$

It is known that the eigenvalues α_i are l-adic units when l is a prime different from p. So, by the product formula, we have

$$\prod_{i=1}^{n!\mathrm{Vol}(\Delta)} |\alpha_i| = q^{\frac{n}{2}n!\mathrm{Vol}(\Delta)}.$$

By a theorem of Kedlaya [9, Theorem 5.6.2], the Frobenius F on $H_0(K_\bullet, \hat{\partial}) \otimes \mathbb{Q}_p[\mu_{q-1}, \pi_m, \pi]$ is of mixed weight $\geq n$. So $q^n F^{-1}$ on $H_0(K_\bullet, \hat{\partial})$ is of mixed weight $\leq 2n - n \leq n$. That is, $|\alpha_i| \leq q^{n/2}$. It follows that all the eigenvalues α_i must have absolute value $q^{n/2}$. This completes the proof of Theorem 1.4.

11 Applications to other situations

Let J be a subset of $\{1, \ldots, n\}$. For $\{j_1, \ldots, j_s\} \subseteq J$, we write

$$B_{\{j_1,\ldots,j_s\}} = \left\{ \sum_{u \in L(\Delta)} a_u x^u \in B : u_{j_1}, \ldots, u_{j_s} > 0 \right\}.$$

For $l = 0, 1, \ldots, n$, we define

$$K_l(f, J) = \bigoplus_{1 \leq i_1 < \cdots < i_l \leq n} B_{J \setminus \{i_1,\ldots,i_l\}} e_{i_1} \wedge \cdots \wedge e_{i_l}.$$

Then $(K_\bullet(f, J), \hat{\partial})$ is a subcomplex of $(K_\bullet(f, \emptyset), \hat{\partial})$. The latter is the complex $(K_\bullet, \hat{\partial})$ we defined earlier.

Lemma 11.1 *The sequence*

$$0 \to K_\bullet(f, J) \to K_\bullet(f, J \setminus \{j\}) \to K_\bullet(f^{\{j\}}, J \setminus \{j\}) \to 0$$

is exact, where $f^{\{j\}}$ is the Witt vector whose i-th coordinate is the sum of monomials of the i-th coordinate of f not divided by x_j.

We define, for $k = 1, 2, \ldots$,

$$S_k(f, J) = \sum_{x^{q^k} = x, x_{i_1} \cdots x_{i_r} \neq 0} \psi\left(\mathrm{Tr}_{\mathbb{Q}_p[\mu_{q^k-1}]/\mathbb{Q}_p}(\omega(f)(x)) \right)$$

if $\{1, \ldots, n\} \setminus J = \{i_1, \ldots, i_r\}$, and $f \in W_m(\mathbb{F}_q[x_1, \ldots, x_n, (x_{i_1} \cdots x_{i_r})^{-1}])$. Here the equation $x^{q^k} = x$ is solved in $(\overline{\mathbb{Q}}_p)^n$. We write

$$L_{f,J}(t) = \exp\left(\sum_{k=1}^{\infty} S_k(f, J) \frac{t^k}{k} \right).$$

By the above lemma we infer the following trace formula from the earlier one.

Proposition 11.2 *For $k = 1, 2, \ldots$, we have*

$$S_k(f, J) = \sum_{l=0}^{n} (-1)^{l+1} Tr\left((p^n F^{-1})^{ak}; H_l(K_\bullet(f, J), \hat{\partial}) \right).$$

Equivalently,

$$L_{f,J}(t) = \prod_{l=0}^{n} \det\left(1 - (p^n F^{-1})^a t; H_l(K_\bullet(f, J), \hat{\partial}) \right)^{(-1)^l}.$$

We call f commode with respect to J if $\Delta_\infty(f)$ is commode with respect to J. Recall that a convex polyhedron Δ in $\mathbb{Q}^n$ that contains the origin is commode with respect to J if it lies in $(\prod_{i=1,i\notin J}^n \mathbb{Q}) \times (\prod_{i\in J} \mathbb{Q}_{\geq 0})$ and $\dim(\Delta_C) = n - |C|$ for all subset C of J, where $\Delta_C = \{(u_1, \ldots, u_n) \in \Delta : u_j = 0 \text{ if } j \in C\}$. By Lemma 11.1 and Proposition 11.2, we infer the following proposition from Theorem 1.3.

Proposition 11.3 *If f is commode with respect to J and non-degenerate with respect to $\Delta_\infty(f)$, then $L_{f,J}(t)$ is a polynomial, and its Newton polygon with respect to ord_q lies above the Hodge polygon of*

$$\sum_{C\subset J}(-1)^{|C|}P_{\Delta_C}\left(t^{\frac{D(\Delta)}{D(\Delta_C)}}\right)$$

with the same endpoints. In particular, $L_{f,J}(t)$ is of degree

$$\sum_{C\subset J}(-1)^{|C|}(n-|C|)!Vol(\Delta_C).$$

From Lemma 10.1 we infer the following one.

Lemma 11.4 *Let Δ be a convex polyhedron in $\mathbb{Q}^n$ of dimension n that contains the origin and is commode with respect to J. Let $(V_{\Delta,J}, U_{\Delta,J})$ be the endpoint of the Hodge polygon of*

$$\sum_{C\subset J}(-1)^{|C|}P_{\Delta_C}\left(t^{\frac{D(\Delta)}{D(\Delta_C)}}\right)$$

other than $(0,0)$. Then

$$U_{\Delta,J} = \frac{n}{2}V_{\Delta,J} + \sum_{l=1}^{|J|+1}(-1)^l\frac{(n-l)!}{2}\left(\sum_{C\subset J,|C|=l-1}S_{\Delta_C} - l\sum_{C\subset J,|C|=l}Vol(\Delta_C)\right).$$

In particular, $U_{\Delta,J} = \frac{n}{2}V_{\Delta,J}$ if the origin is an interior point of Δ_J.

By Lemma 10.3 we infer the following proposition from Proposition 10.2.

Proposition 11.5 *If f is commode with respect to J and non-degenerate with respect to $\Delta := \Delta_\infty(f)$, and the origin lies in the interior of Δ_J, then the reciprocal roots of $L_{f,J}(t)$ are of absolute value $q^{n/2}$.*

Acknowledgment This work is completed when the authors are visiting the Morningside Center of Mathematics in Beijing (MCM). The authors thank MCM for its hospitality, and thank Kezheng Li and Fei Xu for inviting them to visit MCM. The authors also want to express their gratitude to Daqing Wan, who patiently guides this research. Chunlei Liu would like to thank Yingbo Zhang for support, and Lei Fu for discussions. The research of Chunlei Liu is supported by NSFC Grant No. 10371132, by Project 985 of Beijing Normal University, and by the Foundation of Henan Province for Outstanding Youth.

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